Trainings

General Information

The vibration damper controls the twisting or torsional vibration of the crankshaft. A vibration damper is engineered for use on a specific engine model.

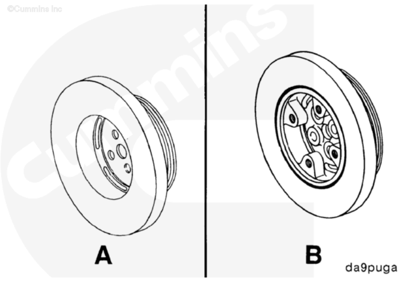
It is **not** economical to repair a vibration damper in the field. Install a new or rebuilt vibration damper if the inspection indicates that a damper is defective.

The viscous vibration damper has a limited service life. The damper **must** be replaced if worn or damaged.

There are two different design vibration dampers used on the B Series engines:

a. Viscous damper (A) for engines rated at speeds above 2500 rpm.
b. Rubber element damper (B) for engines rated at speeds below 2500 rpm.

Note : The rubber vibration damper (B) is available either with or without the crankshaft adapter.

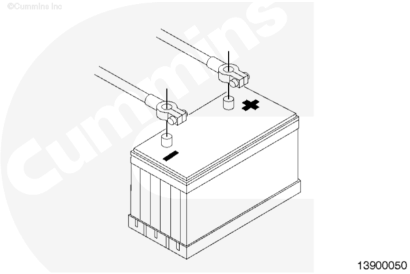


Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries.



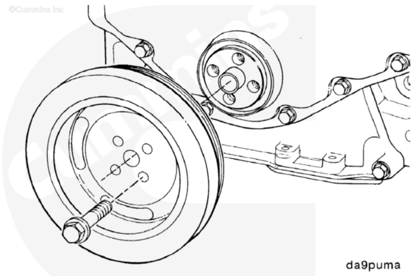
- Remove the drive belt. Refer to Procedure 008-002 (/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html).

Remove

For front gear train engines, remove the four capscrews.

For rear gear train engines, remove the six capscrews.

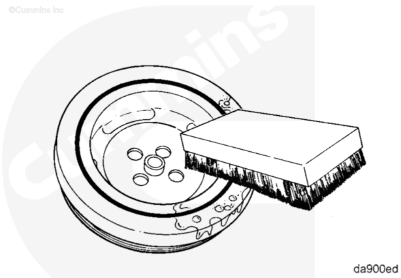
Remove the vibration damper.



Clean and Inspect for Reuse

⚠ WARNING ⚠

Compressed air used for cleaning should not exceed 207 kPa [30 psi]. Wear protective clothing, goggles/shield, and gloves to reduce the possibility of personal injury.

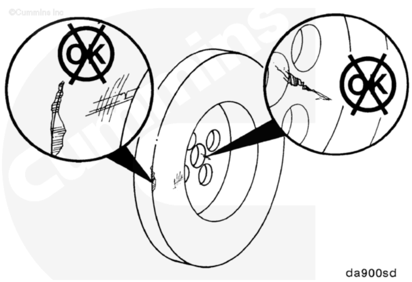


Using soapy water, clean any oil from the vibration damper.

Dry the vibration damper with compressed air.

Check the mounting web for cracks.

Check the alignment marks on the inner and outer rings.

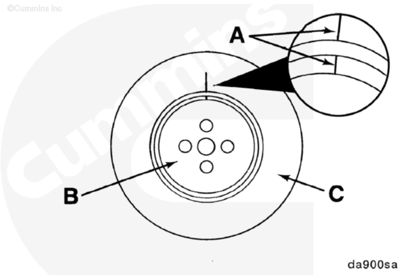


Check the index lines (A) on the damper hub (B) and the inertia member (C).

If the lines are more than 1.59-mm [1/16-in] out of alignment, replace the damper.

Inspect the vibration damper hub for cracks.

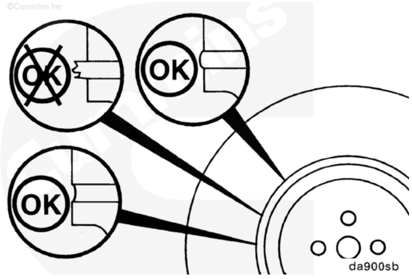
Replace the damper if the hub is cracked.



Inspect the rubber member for deterioration.

If pieces of rubber are missing or if the elastic member is more than 3.18 mm [1/8 in] below the metal surface, replace the vibration damper.

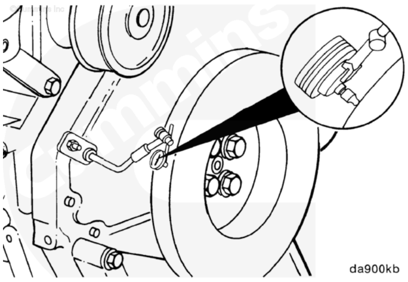
Note : Look for forward movement of the damper ring on the hub. Replace the damper if any movement is detected.



Measure

Measure the vibration dampers eccentricity.

Install a dial indicator as illustrated.

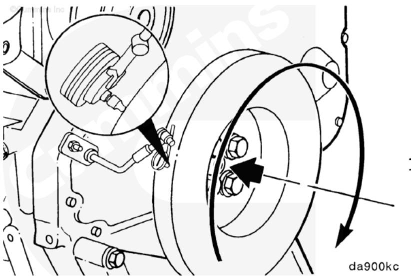


Rotate the crankshaft with engine barring tool, Part Number 3377371.

Record the dial indicators movement.

mm		in
0.10	MAX	0.004

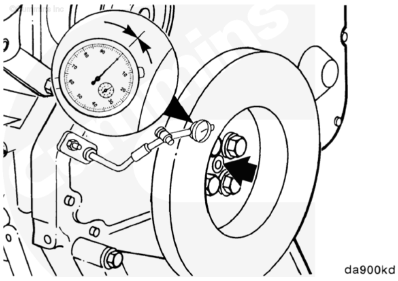
Note : If the eccentricity is not within specification the vibration damper must be replaced.



Measure the vibration damper wobble.

Install a dial indicator as illustrated.

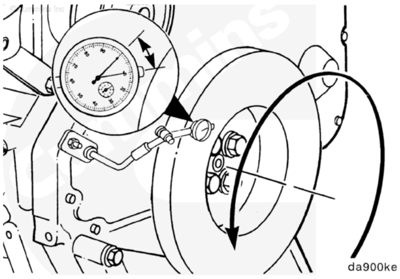
Push the crankshaft to the front or rear and zero the dial indicator.



Rotate the crankshaft with engine barring tool, Part Number 3377371, 360 degrees, maintaining the position of the crankshaft.

Record the dial indicator movement.

mm		in
0.18	MAX	0.007



Install

Front Gear Train

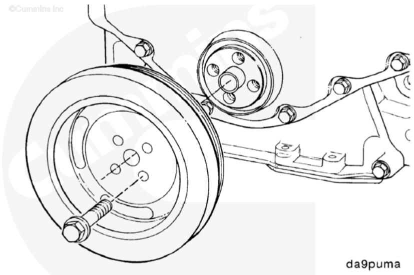
Note : The B Series engines have two configurations for the crankshaft pulleys and vibration dampers. Determine which configuration is used and use the appropriate steps in this procedure.

One Piece Pulley/Vibration Damper

Install the crankshaft vibration damper.

Install and tighten the crankshaft pulley/vibration damper capscrews.

Torque Value: 125 n•m [92 ft-lb]



Two-Piece Pulley/Vibration Damper

Install the vibration damper.

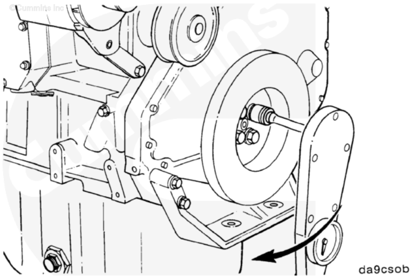
Install and tighten the vibration damper capscrews.

Torque Value: 200 n•m [148 ft-lb]

Install the crankshaft pulley.

Install and tighten the crankshaft pulley capscrews.

Torque Value: 77 n•m [57 ft-lb]



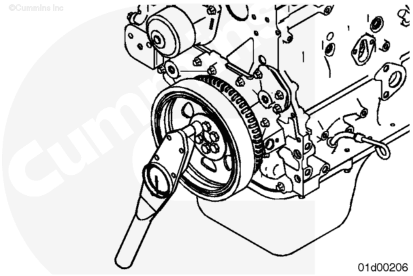
Rear Gear Train

Lubricate bolts with clean engine oil.

Install the vibration damper .

Tighten the six vibration damper capscrews in a criss-cross pattern.

Torque Value:	Step 1	50 n.m [37 ft-lb]
	Step 2	Rotate 90 degrees



Finishing Steps

- Install the drive belt. Refer to Procedure 008-002 (/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html).

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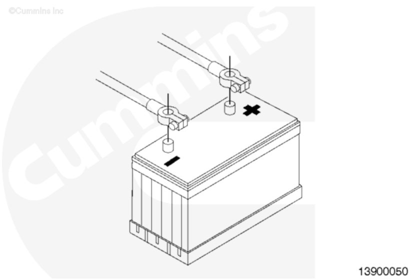

- Connect the batteries.

Last Modified: 31-Mar-2006